

1. Dataset Overview & FAIR Alignment

- **Dataset Name:** FA Women’s Super League (WSL) Spatial PassMap and Markovian Null Ensembles (2023/2024).
- **Short Description:** A curated collection of directed, weighted passing networks and spatial probability tensors derived from the 2023/2024 FA Women’s Super League season, alongside synthetic generative null networks.
- **Source:** Programmatically extracted via the `statsbombpy` API client accessing free StatsBomb open data.
- **Curator/Data Engineering:** Built using a custom Python pipeline (`pandas`, `networkx`, `matplotlib`, `scipy`) for the Network Science assessment.
- **License:** Open-access for academic research under the StatsBomb User Agreement.
- **FAIR Compliance:** Adheres to FAIR principles (Findable, Accessible, Interoperable, Reusable) by converting raw, proprietary spatiotemporal event streams into standardized adjacency matrices, percentage-scaled coordinates, and portable edge-list formats.

2. Motivation & Purpose

- **Primary Purpose:** Created to lay the computational groundwork for domain-aware, spatially constrained generative null models for network analysis in association football.
- **Research Gap Addressed:** Directly responds to systemic underrepresentation of women’s football in quantitative sports literature while addressing the lack of valid null baselines in spatial network science.
- **Intended Use Cases:** Multi-scale network diagnostics (micro-level betweenness centrality, meso-level transitive triad intensity, macro-level shortest path lengths) and benchmarking tactical passing structures.
- **Known Limitations:** Empirical benchmarking across real matches suffers from extreme data sparsity at high pass volumes (> 700 passes). Grouping matches by formation and volume bins yields underpopulated sample sizes, highlighting the necessity of generative null modeling over pure empirical baselines.

3. Dataset Composition & Network Topology

- **Network Format:** Directed, weighted, spatially constrained networks ($G = (V, E, W)$).
- **Nodes (V):** 11 nodes per network, representing the starting 11 tactical positions / active players with the highest appearance duration in each match-team instance.
- **Edges (E) & Weights (W):** Directed pass counts ($A \rightarrow B$) representing successfully completed passes between teammates (incomplete or intercepted passes are filtered out).

- **Macro Corpus Scope:** 264 total team-match networks comprising 105,262 successful completed passes across the 2023/2024 WSL season, filtered to 89,781 for the scope of the project.
- **Case Study Network:** An extreme high-volume match (Arsenal WFC, Match ID 3913160; 847 total passes) utilized for stress-testing spatial null baselines against high-density graph topologies.

4. Data Collection, Normalization & Engineering

- **API Ingestion:** Match lineups and event streams were fetched programmatically using `statsbompy.sb.events()` and `statsbompy.sb.lineups()`.
- **Positional Taxonomy Mapping:** Fine-grained StatsBomb positions (e.g., *Left Center Back*, *Right Center Back*) were mapped into a condensed 11-class taxonomy (e.g., CB, CM, ST) to maximize data density and eliminate team-specific lateral biases during probability modeling.
- **Spatial Normalization & Binning:** Pitch coordinates were normalized from StatsBomb’s raw scale (120×80) to a standardized percentage scale (100×100) and discretized into a 10×10 uniform spatial grid of 100 cells.
- **Markov Probability Tensor:** Compiled a season-wide frequency tensor of pass end locations binned by condensed recipient position. Applied Laplace additive smoothing ($\alpha = 1.0$) to establish a valid conditional probability model $P(\text{Position} \mid \text{Grid Bin})$.
- **Generative Recipient Resampling:** Synthetic null event streams were generated by preserving empirical origin coordinates, passer identities, and total pass counts while resampling recipient positions from $P(\text{Position} \mid \text{Grid Bin})$. Self-passes ($A \rightarrow A$) were strictly prevented via an iterative rejection sampler.

5. Ethical, Privacy & Bias Considerations

- **Privacy & PII:** Contains public-figure PII (player names, positions, and match appearance statistics). As all data is derived from publicly broadcast professional sports events and published open-access databases, it carries zero risk of disclosing sensitive personal, medical, or private life information.
- **Domain Equity:** Promotes open data accessibility in women’s sports science by developing open-source visualization and graph analysis pipelines.
- **Tactical and Performance Biases:** The spatial probability tensor is trained on elite-level professional matchplay (FA WSL). Tactical behaviors (e.g., double-pivot build-ups or terminal off-the-ball striker roles) reflect high-level professional dynamics and may not directly generalize to lower-tier or amateur networks.

6. Suggested Analysis & Benchmarking

- **Recommended Metrics:**

- *Micro*: Unweighted degree ($k_{\text{in}}, k_{\text{out}}$), net flow (Δs_i), and weighted betweenness centrality $g(i)$.
- *Meso*: Pure transitive triad intensity ($\bar{I}_{\text{transitive}}$) isolating progressive wall-pass combinations ($A \rightarrow B \rightarrow C$ with $A \rightarrow C$).
- *Macro*: Global average shortest path length (d_{global}) using inverted weights ($l_{ij} = 1/w_{ij}$) and degree coefficient of variation (CV_k).

- **Null Model Recommendations:** Users should benchmark empirical PassMaps against the provided 1st-order Spatial Markovian null model. Traditional randomizations (e.g., Erdős–Rényi $G(N, p)$ or degree-preserving rewiring) are fundamentally invalid for football networks because they destroy pitch geometry, flatten workload inequality, and generate impossible tactical anomalies (such as high-volume passing triangles between goalkeepers and strikers).